

# Dirichlet's Theorem

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# Main Theorem Prerequisites

## Dirichlet's Theorem on Arithmetic Progression

Every Arithmetic Progression with first term  $a$  and common difference  $q$ , such that  $\gcd(a, q) = 1$ , contains infinitely many primes.

Some useful theorems from Complex Analysis

## Theorem 1

If  $f_n(z)$  is analytic  $\text{seq}^n$  in  $\Omega_n$  and  $\{f_n(z)\} \rightarrow f(z)$  in  $\Omega$  uniformly on every compact set of  $\Omega$ . Then  $f(z)$  is analytic in  $\Omega$ . Moreover  $f'_n(z) \rightarrow f'(z)$  on every cpt sets of  $\Omega$

## Theorem 2

Suppose  $\{a_n\}$  is a complex sequence. Then  $\prod_{n=1}^{\infty} (1 + a_n)$  with  $1 + a_n \neq 0$  converges (absolutely) simultaneously with the series  $\sum_{n=1}^{\infty} \log(1 + a_n)$  whose terms are values in principal branch of logarithm.

## Theorem 3

Let  $a_1, a_2, \dots, a_n \in \mathbb{C}$  be a sequence s.t  $1 + a_n \neq 0$  then  $\prod_{n=1}^{\infty} (1 + a_n)$  converges absolutely iff  $\sum_{n=1}^{\infty} |a_n|$  converges absolutely.

## Theorem 4

Let  $a_1, \dots, a_n \in \mathbb{C}$ ,  $|a_i| < 1 \forall i$  and  $\sum_{i=1}^{\infty} |a_i| < \infty$ . Then

$$\prod_{i=1}^{\infty} (1 - a_i)^{-1} = 1 + \sum_{j=1}^{\infty} \sum_{(r_1, \dots, r_j)} a_1^{r_1} \dots a_j^{r_j}$$

The sum and product are both absolutely convergent, hence independent of the order of  $a_i$ .

# Euler's Product

We begin with Euler's proof of existence of infinitely many primes using the product form of Riemann's Zeta Function called Euler Product.

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1} \quad (1)$$

where the product runs over all the primes  $p$ . The proof of the above equality follows from expanding the product into geometrical series for  $s > 1$ , exchanging terms which is allowed by Fubini's Theorem as all terms are positive and finally using Fundamental Theorem of Arithmetic. Taking log and expanding, we have,

$$\log \zeta(s) = \sum_p \sum_{m \in \mathbb{N}} m^{-1} p^{-ms} \quad (2)$$

Now, since  $\zeta(s) \rightarrow \infty$  as  $s \rightarrow 1$ , and

$$\sum_p \sum_{m=2}^{\infty} m^{-1} p^{-ms} < \sum_p \sum_{m=1}^{\infty} p^{-m} = \sum_p \frac{1}{p(p-1)} < 1, \quad (3)$$

which is bounded. So  $\sum_p p^{-s}$  has to diverge, i.e.  $\sum_p p^{-s} \rightarrow \infty$  as  $s \rightarrow 1$ .  
 If there is finitely many primes, the sum of their reciprocals can't be infinity. Thus we conclude, there are infinitely many primes.

# Dirichlet Characters

We define character function  $\chi$  as a group homomorphism from a finite abelian group to roots of unity in  $\mathbb{C}$ . Dirichlet character modulo  $m$  function is constructed using the above functions only.

For the given prime  $q$ , let  $\omega$  be the  $q-1^{\text{th}}$  root of unity, i.e.

$$\omega^{q-1} = 1$$

Let  $\nu(n)$  be the index of a fixed primitive root, say  $g$  of  $n$  modulo  $q$ , i.e.

$$g^{\nu(n)} \equiv n \pmod{q}$$

We claim that for every  $n$ ,  $\nu(n)$  is uniquely determined.

If possible, for a given  $n$  relatively prime to  $q$ ,  $\exists$  distinct indices  $\nu_1$  and  $\nu_2$ , such that

$$\begin{aligned} g^{\nu_1} &\equiv n \equiv g^{\nu_2} \pmod{q} \\ \Rightarrow g^{\nu_1} &\equiv g^{\nu_2} \pmod{q} \\ \Rightarrow \nu_1 &\equiv \nu_2 \pmod{q-1}, \end{aligned}$$

# Construction

since,  $\phi(q) = q - 1$  is the order of primitive root  $g$ . Thus,  $\nu(n)$  is unique, for every  $n$  relatively prime to  $q$ , modulo  $q - 1$ .

Now we define,  $\chi(n) : \mathbb{N} \rightarrow \mathbb{C}$  such that

$$\chi(n) = \begin{cases} \omega^{\nu(n)}, & \text{if } \gcd(n, q) = 1 \\ 0, & \text{if } \gcd(n, q) > 1 \end{cases}$$

So basically what we do is reduce  $\mathbb{Z}$  to  $U(\mathbb{Z}/m\mathbb{Z})$  and then map it to  $\phi(m)$  roots of unity in  $\mathbb{C}$ . Now we will construct the general case as  $U(\mathbb{Z}/m\mathbb{Z})$  has different structures depending on powers of 2 and primes in  $m$ .

**Case-I:**  $q = p^\alpha$ ,  $p$  is an odd prime and  $\alpha \in \mathbb{Z}^+$ . A primitive root modulo  $q$  exists and has an order of  $\phi(q) = p^{\alpha-1}(p-1)$ . Character modulo  $q$  is given by  $\chi : \mathbb{N} \rightarrow \mathbb{C}$  defined as

$$\chi(n) = \begin{cases} \omega^{\nu(n)}, & \text{when } (n, q) = 1 \\ 0, & \text{when } (n, q) > 1, \end{cases}$$

where  $\omega$  is  $\phi(q)^{th}$  root of unity, and  $\nu$  gives the index  $k$  of a fixed primitive root  $g$  modulo  $q$  such that  $g^k \equiv n \pmod{q}$ .

# Construction(contd.)

## Case-II: $q = 4$

Here, 3 is a primitive root and the character is principal for  $\omega = 1$ . For  $\omega = -1$ , we have

$$\chi(n) = \begin{cases} 1, & \text{when } n \equiv 1 \pmod{4} \\ -1, & \text{when } n \equiv -1 \pmod{4} \\ 0, & \text{when } 2|n, \end{cases}$$

## Case-III: $q = 2^\alpha$ , where $\alpha > 2$ is a positive integer.

Here Order of 5 modulo  $2^\alpha$  is  $2^{\alpha-2} = \frac{1}{2}\phi(2^\alpha)$  and  $5^\alpha \equiv 1 \pmod{4}$ . So  $(-1)^l 5^k$  generates  $U(\frac{\mathbb{Z}}{2^\alpha \mathbb{Z}})$ . Character modulo  $q$  is given by

$$\chi(n) = \begin{cases} \omega^{\nu(n)} \omega'^{\nu'(n)}, & \text{when } (n, q) = 1 \\ 0, & \text{when } (n, q) > 1, \end{cases}$$

where  $\omega$  is square root of unity and  $\omega'$  is  $2^{\alpha-2}$ th root of unity.  $\nu'$  gives the index  $k$ , such that  $(-1)^l 5^k \equiv n \pmod{q}$  and  $\nu = l = 0, 1$ .

# Construction

By Fundamental Theorem of Arithmetic, any positive integer  $q$  can be written as,  $q = 2^{\alpha_0} p_1^{\alpha_1} p_2^{\alpha_2} p_3^{\alpha_3} \dots$ , for suitable choices of  $\alpha_i$ 's.

**Case-IV:** The general character for  $q$  is now given by

$$\chi(n) = \begin{cases} \chi(n; 2^{\alpha_0}) \chi(n; p_1^{\alpha_1}) \chi(n; p_2^{\alpha_2}) \dots, & \text{when } (n, q) = 1 \\ 0, & \text{when } (n, q) > 1, \end{cases}$$

This has this type of structure as  $U(\mathbb{Z}/q\mathbb{Z}) \cong U(\frac{\mathbb{Z}}{2^{\alpha_0}\mathbb{Z}}) \times U(\frac{\mathbb{Z}}{p_1^{\alpha_1}}) \dots \times U(\frac{\mathbb{Z}}{p_n^{\alpha_n}})$

Number of characters modulo  $q$  is  $\phi(2^{\alpha_0}) \phi(p_1^{\alpha_1}) \phi(p_2^{\alpha_2}) \dots = \phi(q)$ .

We represent the characters in terms of exponentials.

$$\chi(n; p^\alpha) = e\left[\frac{m\nu}{\phi(p^\alpha)}\right] \quad \chi(n; 2^\alpha) = e\left[\frac{m\nu}{2} + \frac{m'\nu'}{\phi(2^{\alpha-2})}\right]$$

$$\chi(n) = e\left[\frac{m\nu}{2} + \frac{m'\nu'}{2^{\alpha_0-2}} + \frac{m_1\nu_1}{p_1^{\alpha_1}} + \frac{m_2\nu_2}{p_2^{\alpha_2}} + \dots\right]$$

$m_i$ 's depend on the choice of  $\chi$  and  $\nu_i$ 's

# Properties of $\chi$

## Property of Linear Combination

$$\sum_n \chi(n) = \begin{cases} \phi(q), & \text{when } \chi = \chi_0 \\ 0, & \text{otherwise} \end{cases}$$

$$\sum_{\chi} \chi(n) = \begin{cases} \phi(q), & \text{when } n \equiv 1 \pmod{q} \\ 0, & \text{otherwise} \end{cases}$$

$$\frac{1}{\phi(q)} \sum_{\chi} \bar{\chi}(a) \chi(n) = \begin{cases} 1, & \text{when } n \equiv a \pmod{q} \\ 0, & \text{otherwise} \end{cases}$$

# Dirichlet's L Function

We define  $L$ -functions for  $\chi$  that is character modulo  $m$

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} \quad (4)$$

where for  $s > 1$  it converges absolutely and uniformly for  $s \geq \delta > 0$ . Hence by Theorem 1 it is analytic on  $s > 1$ . Since  $\chi(p)$  is multiplicative,  $L(s, \chi)$  can be re-written in the Euler's Product form by applying Theorem 4 as,

$$L(s, \chi) = \prod_{p \nmid q} \left(1 - \frac{\chi(p)}{p^s}\right)^{-1} \quad \text{for } s > 1 \quad (5)$$

Here  $|\chi(p)p^{-s}| \leq \frac{1}{2}$  for all primes  $p$  and  $s > 1$ . Hence,

$$\frac{2}{3} < \frac{1}{|1 + |\chi(p)p^{-s}||} < |(1 - \chi(p)p^{-s})^{-1}|$$

Thus,  $L(s, \chi)$  does not vanish  $\forall s > 1$ .

We observe that, as  $s \rightarrow \infty$ ,  $L(s, \chi) \rightarrow 1$ , which implies  $\log(L(s, \chi)) \rightarrow 0$ . Taking log on both sides of (5) and using logarithmic expansion, we have,

$$\log(L(s, \chi)) = \sum_{p \nmid q} \sum_{m \in \mathbb{N}} m^{-1} \chi(p^m) p^{-ms} \quad (6)$$

Here, we consider the principal branch value of  $\log(L(s, \chi))$  as the right hand side is continuous on  $s > 1$ . Multiplying (6) by  $\chi^{-1}(a)$ , summing over all such  $\chi$  and using the properties, we have,

$$\sum_{\chi} \overline{\chi(a)} \log(L(s, \chi)) = (\phi(q)) \sum_{p, p^m \equiv a \pmod{q}} \sum_{m \in \mathbb{N}} m^{-1} p^{-ms} \quad (7)$$

$$\Rightarrow \frac{1}{\phi(q)} \sum_{\chi} \overline{\chi(a)} \log(L(s, \chi)) = \sum_{p, p^m \equiv a \pmod{q}} \sum_{m \in \mathbb{N}} m^{-1} p^{-ms} \quad (8)$$

Since for  $m > 1$  the sum on the right hand side of (8) is bounded as shown in (3)

Thus,

$$\frac{1}{\phi(q)} \sum_{\chi} \overline{\chi(a)} \log(L(s, \chi)) < \sum_{p \equiv a \pmod{q}} p^{-s} + 1 \quad (9)$$

Thus proving the left hand side of (9) to be diverging to infinity as  $s \rightarrow 1$  from right will in turn prove the statement of the theorem.

Now in the expression  $\sum_{\chi} \overline{\chi(a)} \log(L(s, \chi))$ , there is a term where  $\chi_0 = 1$ . In this particular case,

$$L(s, \chi_0) = \sum_{n \in \mathbb{N} \atop (q, n)=1} n^{-s} = \prod_{p \nmid q} (1 - p^{-s})^{-1} = \left( \prod_{p|q} (1 - q^{-s}) \right) \zeta(s), \quad (10)$$

which diverges as  $s \rightarrow 1$  from right. Thus,  $\log(L(s, \chi_0))$  goes to  $+\infty$  as  $s \rightarrow 1$  from right. Now, if we can show that for all the other  $\chi \neq 1$ ,  $\log(L(s, \chi))$  is bounded as  $s \rightarrow 1$  from right, we will be done with our proof.

In  $L(s, \chi)$  the sequence  $n^{-s}$  is monotone and convergent to 0 and  $\sum_{n=1}^M \chi(n)$  is bounded for any  $M \in \mathbb{N}$  due to the property of nontrivial  $\chi$ . From Dirichlet's test of uniform convergence of series, we see that  $L(s, \chi)$  converges uniformly  $\forall s \geq \delta > 0$ . Hence,  $L(s, \chi)$  is continuous  $\forall s > 0$ . Thus, proving  $L(s, \chi) \neq 0$  will suffice to show that  $\log(L(s, \chi))$  is bounded as  $s \rightarrow 1$  from right. We now separate the following part of the proof into two cases  $\chi$  is complex and  $\chi$  is real, i.e.  $\chi = -1$  (generator mapsto -1.)

## Case-1: $\chi$ is complex

Putting  $a = 1$  in (8) we get  $\chi(a) = 1 \forall \chi$ , from which we have,

$$\frac{1}{\phi(q)} \sum_{\omega} \log(L_{\omega}(s)) = \sum_{p, p^m \equiv 1 \pmod{q}} \sum_{m \in \mathbb{N}} m^{-1} p^{-ms} \quad (11)$$

Every term on the right hand side of (11) is positive, so we get that  $\sum_{\chi} \log(L(s, \chi)) \geq 0$ , i.e.  $\prod_{\chi} L(s, \chi) \geq 1 \forall s > 1$ .

From (10) and for  $1 < s < 2$ , we have

$$L(s, \chi_0) = \prod_{p|q} (1 - q^{-s}) \zeta(s) < \prod_{p|q} (1 - q^{-2}) \zeta(s) \quad (12)$$

$$\zeta(s) = \sum_{n \in \mathbb{N}} n^{-s} < 1 + \int_1^{\infty} \frac{1}{x^s} dx = 1 + \frac{1}{1-s} = \frac{s}{s-1} < \frac{2}{s-1} \quad (13)$$

So, combining (12) and (13), we get,

$$L(s, \chi_0) < \prod_{p|q} (1 - q^{-2}) \frac{2}{s-1} = \frac{K}{s-1}, \quad (14)$$

where,  $K = \prod_{p|q} 2(1 - q^{-2})$  is independent of  $s$ . Let  $\exists \chi$  such that  $L(1, \chi) = 0$ . Then  $L(1, \bar{\chi}) = 0$  and for other nontrivial characters the limit of Lfunctions is bounded on  $s = 1$  as they are continuous say  $K$ .

Now we have,

$$L(s, \chi) = L(s, \chi) - L(1, \chi) \quad (15)$$

Since  $L(s, \chi)$  is continuous and differentiable for  $s > 1$ , by Mean Value Theorem,

$$\frac{L(s, \chi) - L(1, \chi)}{s - 1} = L'(s_1, \chi), \quad (16)$$

where  $s_1 \in (1, s)$  and  $L'(s, \chi) = - \sum_{n \in \mathbb{N}} (q, n) = 1 \chi(n) \log(n) n^{-s}$ . By Dirichlet's test again, we get that  $L'(s, \chi)$  is continuous  $\forall s > 0$ . Hence,  $|L'(s, \chi)|$  is bounded  $\forall s > 1$  (say  $K_1$ ) as it is decreasing function and thus we get,

$$|L(s, \chi)| < K_1(s - 1), \quad (17)$$

where,  $K_1$  is independent of  $s$ . Thus, taking limit  $s \rightarrow 1$ , we have  $|L(s, \chi)| \rightarrow 0$ . The same holds for  $\bar{\chi}$ .

Hence, as  $s \rightarrow 1$  from right,  $\prod_{\chi} L(s, \chi) \leq KK_1K_2(s-1) \rightarrow 0$ . This is a contradiction to the fact that  $\prod_{\chi} L(s, \chi) \geq 1 \forall s > 1$ . Thus,  $\nexists$  any complex  $\chi$  such that  $L(1, \chi) = 0$

**Case 2:**  $\chi = -1$

### Result on $\zeta(s)$

The zeta function  $\zeta : \mathbb{C} \rightarrow \mathbb{C}$  defined as

$$\zeta(s) = \sum_{n \in \mathbb{N}} n^{-s} \quad (18)$$

is meromorphic for  $\sigma > 0$ , where  $s = \sigma + it$  with a simple pole at  $s = 1$  with Residue  $\text{Res}(\zeta, 1) = 1$ .

Using equation (10) and the previous result, we see that,

$$\text{Res}(L_{\chi_0}, 1) = \lim_{s \rightarrow 1} (s-1)\zeta(s) \prod_{p|q} (1 - p^{-s}) = \frac{\phi(q)}{q} \quad (19)$$

For non-principal real characters, we define a sum  $S : \mathbb{R} \rightarrow \mathbb{C}$  as  $S(x) = \sum_{n \leq x} \chi(n)$ . Then,

$$L(s, \chi) = \sum_n \chi(n) n^{-s} = \sum_n S(n) [n^{-s} - (n+1)^{-s}] \quad (20)$$

$$= \sum_n s \int_n^{n+1} S(x) x^{-s-1} dx = s \int_1^\infty S(x) x^{-s-1} dx \quad (21)$$

As  $S(x)$  is bounded, the integral on R.H.S. is bounded and shows  $L(s, \chi)$  is absolutely convgt. for  $x > 0$  and converge uniformly for  $x \geq \delta > 0$  and so by 1st Theorem it is analytically continuable as a regular function to  $x > 0$ . If possible, we assume for a non-principal real character let  $L(1, \chi) = 0$ . This implies that  $L(s, \chi)L(s, \chi_0)$  is analytic for  $\sigma > 0$ . Also,  $L(2s, \chi_0)$  is analytic for  $\sigma > \frac{1}{2}$  and diverges as  $s \rightarrow \frac{1}{2}^+$ .

Define  $\psi : \mathbb{C} \rightarrow \mathbb{C}$  as

$$\psi(s) = \frac{L(s, \chi)L(s, \chi_0)}{L(2s, \chi_0)} \quad (22)$$

As numerator is bounded,  $\psi(s) \rightarrow 0$  as  $s \rightarrow \frac{1}{2}^+$ . Now by the assumption,  $\chi(p) = \pm 1$ ,  $p \nmid q$ . The Euler product term corresponding to  $p$ 's with  $\chi(p) = -1$  is

$$\frac{(1 + p^{-s})^{-1}(1 - p^{-s})^{-1}}{(1 - p^{-2s})^{-1}} = 1 \quad (23)$$

Thus,

$$\psi(s) = \prod_{\chi(p)=1} \left( \frac{1 + p^{-s}}{1 - p^{-s}} \right) \quad \text{for } \sigma > 1 \quad (24)$$

Let  $\sum_n a_n n^{-s}$  be the corresponding Dirichlet Series for  $\psi$  which is true since every euler product has a corresponding dirichlet series. This valid for  $\sigma > 1$

$\psi$  is analytic for  $\sigma > \frac{1}{2}$ . So,

$$\psi(s) = \sum_m \frac{1}{m!} \psi^{(m)}(2)(s-2)^m, \quad (25)$$

which is a power series centred at 2 and with radius of convergence  $\frac{3}{2}$ . Hence, by the Dirichlet Series,

$$\psi^{(m)}(2) = (-1)^m \sum_n a_n (\log(n))^m n^{-2} = (-1)^m b_m (\text{say}) \quad (26)$$

$$\Rightarrow \psi(s) = \sum_{m=0}^{\infty} \frac{1}{m!} b_m (2-s)^m, \quad (27)$$

where  $b_m \geq 0$  and  $s \in (\frac{1}{2}, \frac{7}{2})$ . As  $\psi \geq 0$  for  $s < 2$  and  $a_1 = 1$ ,  $\psi(s) \geq \psi(2) = b_0 = \sum_{n=1}^{\infty} \frac{a_n}{n^2} \geq 1$ , a contradiction to continuity  $\psi(s) \rightarrow 0$  as  $s \rightarrow \frac{1}{2}^+$ . Thus  $L(1, \chi) = 0$  is disproved.

- Multiplicative Number Theory by Davenport
- A Classical Introduction to Modern Number Theory by Ireland and Rosen